

REPORT_Project Review 1

TEAM NO: 8

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Q&A:

1. What is an agent and an endpoint in our project?

Answer:

In our system, an agent is a DQN-based controller assigned to a traffic intersection. It observes the current state (like queue length, waiting times, and signal phase) and decides the best action (whether to switch lights or keep them). An endpoint refers to the IoV devices that collect and transmit data for example, vehicles, roadside units (RSUs), or the traffic signal controllers themselves. Agents use this endpoint data to make real-time decisions.

2. Why do we need edge computing when we already have normal threshold-based traffic systems?

Answer:

Threshold-based systems work on simple rules (like “if queue > 10, extend green”), but they cannot adapt well to unexpected changes such as accidents, sudden traffic surges, or city-wide congestion. Centralized cloud systems could adapt, but they introduce high latency and processing bottlenecks. Edge computing allows decisions to be made closer to the intersections (at ES1/ES2), reducing delay, scaling across the whole city, and ensuring real-time

adaptive control. Our paper shows that DRLE achieves better convergence, scalability, and efficiency compared to static or centralized approaches.

3. How exactly are we going to implement all of this?

Answer:

We will implement DRLE in a simulated environment. SUMO will simulate real-world traffic networks, while Flow integrates SUMO with RLlib, which runs our DQN agents at each intersection. To make communication realistic, we'll bring in NS-3 to simulate V2X network delays. The entire setup is orchestrated using Python, representing ES1 and ES2 edge servers. The output will be seen in SUMO's GUI (vehicle movement, signal control) and measured with metrics such as average waiting time, queue length, and learning speed.

4. Can you explain the intersection architecture and the overall DRLE system design?

Answer:

At the intersection level (ES1), each traffic light has a control unit with sensors or cameras to collect real-time data. This data is sent to a nearby edge server (ES1), which hosts the local DQN agent controlling that intersection. At the regional level (ES2), data from multiple ES1 servers is aggregated to ensure coordinated decisions across a city region. The ES2 servers can also send backups to the cloud for long-term storage or analysis. This three-level architecture (device layer → ES1 → ES2) allows decentralized decision-making with global coordination, making the system both scalable and resilient.

Final Summary

Our paper proposes Decentralized Reinforcement Learning at the Edge (DRLE) as a solution to urban traffic congestion. Unlike traditional fixed-time or centralized systems, DRLE places intelligent agents at the edge servers (ES1, ES2), allowing intersections to make real-time, low-latency decisions using

data from the Internet of Vehicles (IoV). Each agent is trained with Deep Q-Networks (DQN) to adapt traffic signals dynamically, while the multi-level edge hierarchy ensures coordination across a city.

We plan to implement this in a simulated environment using SUMO for traffic, Flow + RLlib for reinforcement learning, and NS-3 for network delay modeling. The system will be validated by measuring performance metrics like waiting times, queue lengths, and vehicle speed across different grid sizes.

Overall, DRLE provides a scalable, adaptive, and realistic approach to traffic management bridging the gap between theory and smart city deployment.